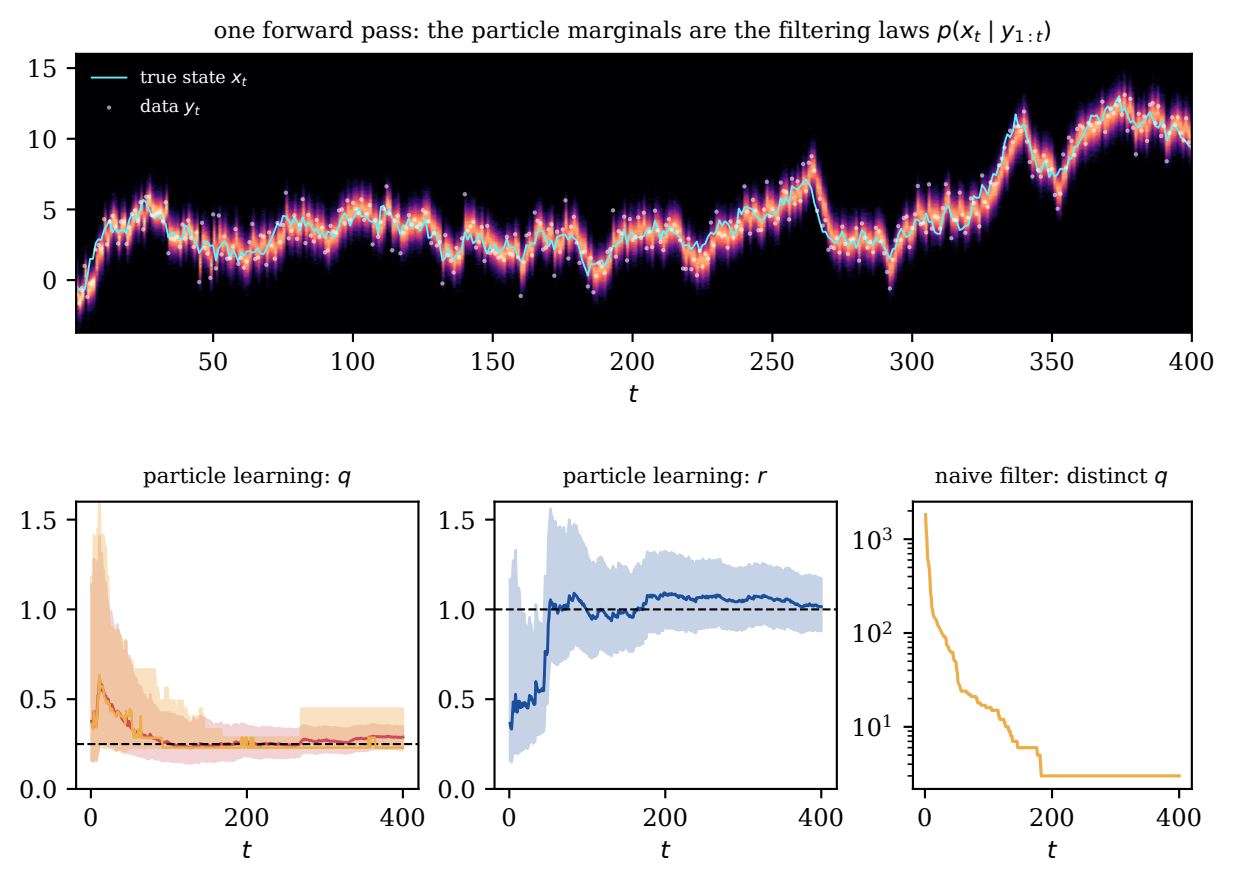

one forward pass: the particle marginals are the filtering laws $p(x_t \mid y_{1:t})$
true state $x_t$
data $y_t$
15
10
5
0
50
100
150
200
250
300
350
400
$t$
particle learning: $q$
particle learning: $r$
naive filter: distinct $q$
1.5
1.0
0.5
0.0
$10^3$
$10^2$
$10^1$
0
200
400
$t$